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Comptia N+ Assignment

[N+ - Network security, Maintenance and Troubleshooting procedures]

Network Troubleshooting, Management, Monitoring, and Optimization

Task:

❖ **1. Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.**

➢ **Aim**

- To use the ipconfig /all command in Windows Command Prompt to identify the computer's IPv4 address, subnet mask, default gateway, and DNS server addresses.

➢ **Introduction**

- **IP Configuration** is an important part of computer networking. Every device connected to a network requires network configuration to communicate with other devices and access the Internet.
- Windows provides a command-line utility called **ipconfig**, which displays the IP configuration of network adapters.

- The ipconfig /all command provides detailed information about network adapters, including IPv4 address, IPv6 address, subnet mask, default gateway, DNS servers, physical address (MAC address), and DHCP information.
- In this practical, the ipconfig /all command is used to examine the network configuration of a Windows computer.

➢ **Requirements**

- A Windows computer or laptop
- An active Wi-Fi or Ethernet connection
- Windows Command Prompt

➢ **Procedure**

➢ **Step 1: Open Command Prompt**

1. Press Windows + R on the keyboard.
2. Type cmd in the Run dialog box.
3. Press Enter to open Command Prompt.

➢ **Step 2: Execute the Command**

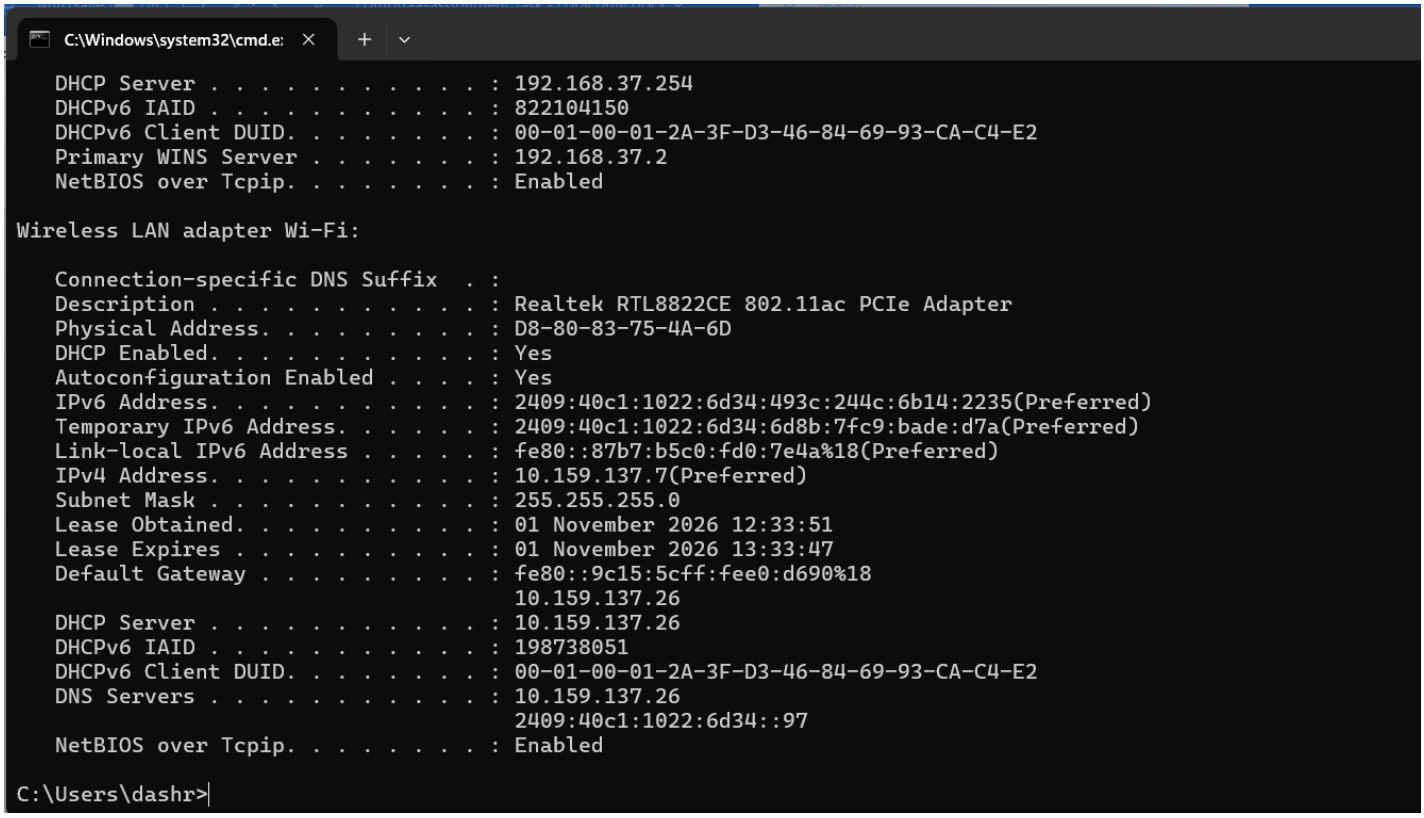
1. Type the following command:
   ipconfig /all
2. Press Enter to execute the command.

➢ **Step 3: Identify the Active Network Adapter**

1. Find the active network adapter, such as Wireless LAN adapter Wi-Fi or Ethernet adapter.
2. Check the network details displayed under that adapter.

3. Identify the IPv4 Address, Subnet Mask, Default Gateway, and DNS Servers.

➢ **Step 4: Record the Network Configuration**



➢ **Observation**

➢ The following table should be completed using the actual output of the ipconfig /all command.

Sr. No.	Network Parameter	Observed Value
1	IPv4 Address	10.159.137.7

2	Subnet Mask	255.255.255.0
3	Default Gateway	10.159.137.26
4	DNS Server(IPv4)	10.159.137.26

- **Explanation of Network Parameters**
  1. **IPv4 Address:** A logical address assigned to a device to identify it on an IPv4 network.
  2. **Subnet Mask:** Identifies the network portion and host portion of an IPv4 address.
  3. **Default Gateway:** The router address used by a computer to communicate with devices outside its local network.
  4. **DNS Server:** A server that translates domain names, such as www.google.com, into IP addresses.

- **Result**
- The ipconfig /all command was successfully executed in Windows Command Prompt. The IPv4 address, subnet mask, default gateway, and DNS server addresses were identified from the network configuration.

❖ **2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.**

➢ **Aim**

- To simulate a network connectivity issue by disconnecting and reconnecting Wi-Fi, use the ping command to test connectivity to Google, and compare the results.

➢ **Introduction**

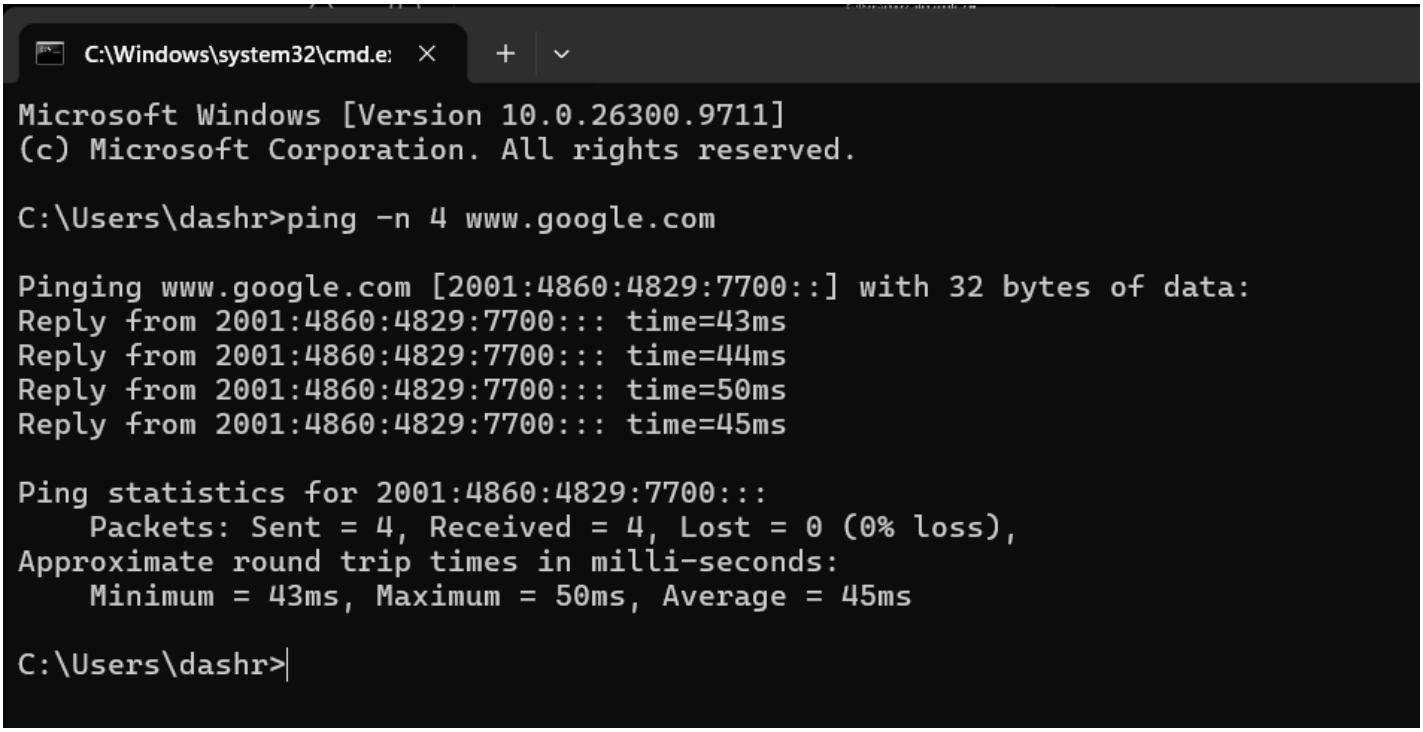
- Network connectivity is essential for communication between computers and Internet servers. Sometimes, a computer may lose its Internet connection because of a disconnected Wi-Fi network or Ethernet cable.
- The **ping** command is a network troubleshooting utility used to check whether a destination is reachable. It sends ICMP Echo Request packets and waits for replies from the destination.
- In this practical, the ping command is used before and after reconnecting the network to observe the difference in connectivity results.

➢ **Requirements**

- A Windows computer or laptop
- Wi-Fi or Ethernet connection
- Windows Command Prompt
- Internet access

- **Procedure**
- **Step 1: Test Connectivity Before Disconnecting**

1. Connect the computer to Wi-Fi or Ethernet.
2. Open Command Prompt.
3. Execute the following command:
   ping -n 4 www.google.com
4. Observe the replies and record the results.

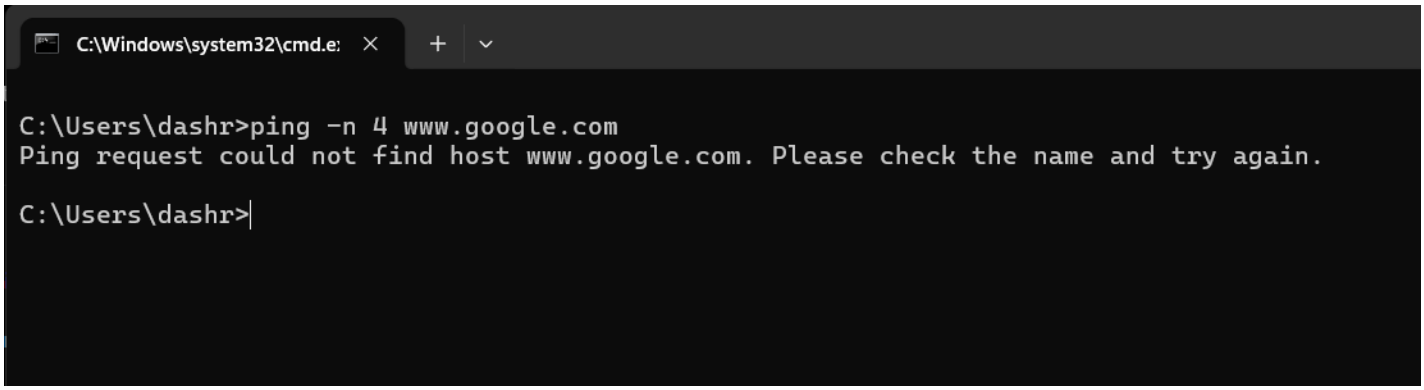


- **Step 2: Disconnect the Network**

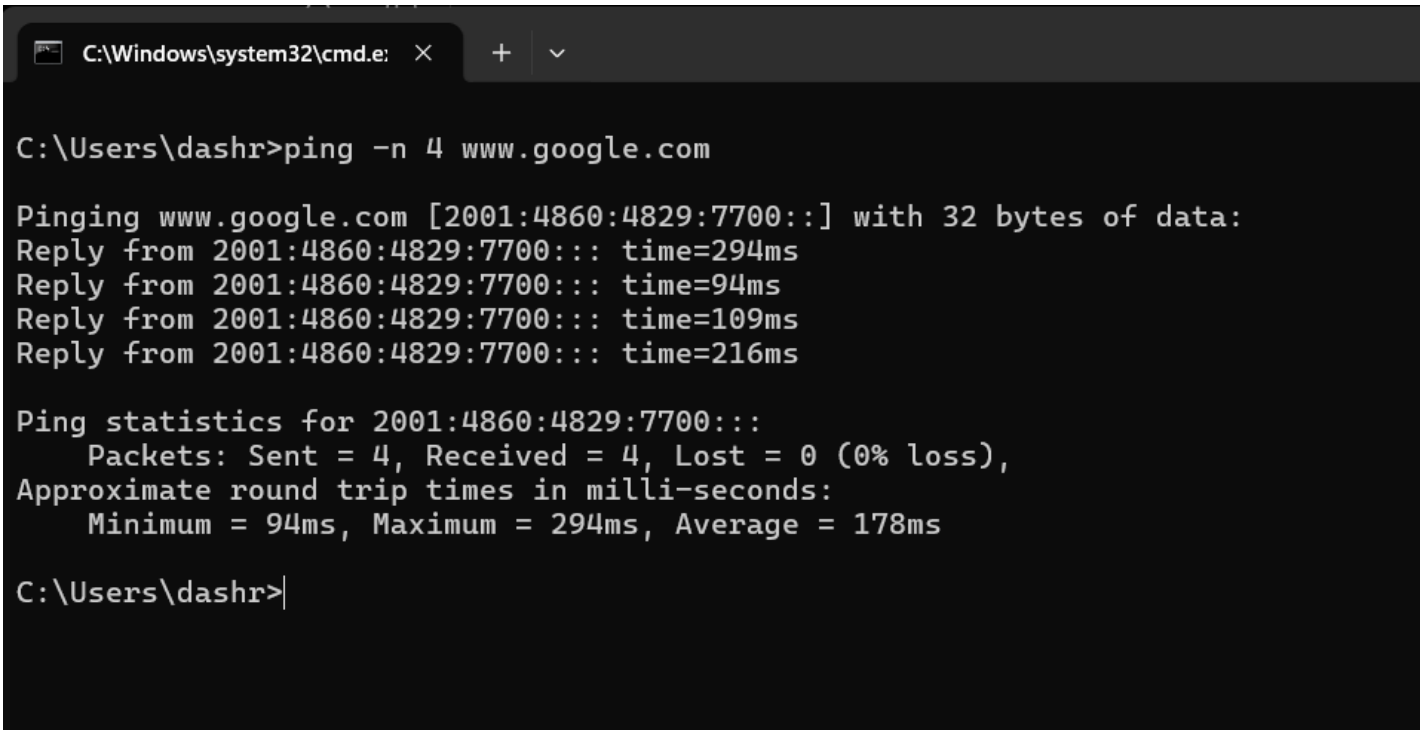
1. Disconnect Wi-Fi using the Windows network settings, or safely unplug the Ethernet cable.
2. Open Command Prompt.
3. Execute the same ping command:

ping -n 4 www.google.com

4. Observe the output and record the result.



- **Step 3: Reconnect the Network**
  1. Reconnect the computer to the same Wi-Fi network or Ethernet connection.
  2. Wait until the network connection is restored.
  3. Execute the ping command again:
     ping -n 4 www.google.com
  4. Observe the replies and record the response times and packet loss.



- **Observation**
- The following table shows the expected difference in results. Replace the status with the actual output observed during the practical.

Network Condition	Ping Result	Average Response Time	Status
Before Disconnection	4 packets received, 0% packet loss	45 ms	Connected
During Disconnection	Could not find host www.google.com	Not Available	Disconnected

After Reconnection	Pending actual test	Pending	Pending

- **Explanation of Ping Status**
  1. **Reply from:** Indicates that a response was received from the destination, showing that it is reachable.
  2. **Request timed out:** Indicates that no reply was received within the allowed time.
  3. **Destination host unreachable:** Indicates that the network could not find a route to the destination.
  4. **Packet Loss:** Shows the percentage of packets sent without receiving a reply.

- **Difference Between Results**
- When the computer is connected to the Internet, the ping command normally receives replies from Google. After disconnecting Wi-Fi or Ethernet, the computer may show request timeouts or destination unreachable messages because network connectivity is unavailable.
  - After reconnecting, ping replies should normally return, showing that network connectivity has been restored.

- **Result**
- The ping command was used to test connectivity to www.google.com before disconnection, during disconnection, and after reconnection. The results demonstrated the effect of network disconnection on Internet connectivity.

❖ 3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

➢ Aim

- To monitor network usage for five minutes while streaming a YouTube video using Windows Task Manager or Resource Monitor, record the peak network usage, and identify the process that uses the most bandwidth.

➢ Introduction

- Network monitoring is the process of observing network activity and measuring the amount of data transmitted and received by a computer.
- Windows provides built-in tools such as **Task Manager** and **Resource Monitor** to monitor system performance and network activity.
- Task Manager displays network usage by running applications and processes. Resource Monitor provides more detailed information about network activity, including data transfer by individual processes.
- In this practical, network usage is monitored while streaming a YouTube video for five minutes. The peak network usage and the process using the most bandwidth are recorded.

➢ Requirements

- A Windows computer or laptop

- An active Internet connection
- A web browser
- YouTube video
- Windows Task Manager or Resource Monitor

➢ **Procedure**

➢ **Step 1: Open Task Manager**

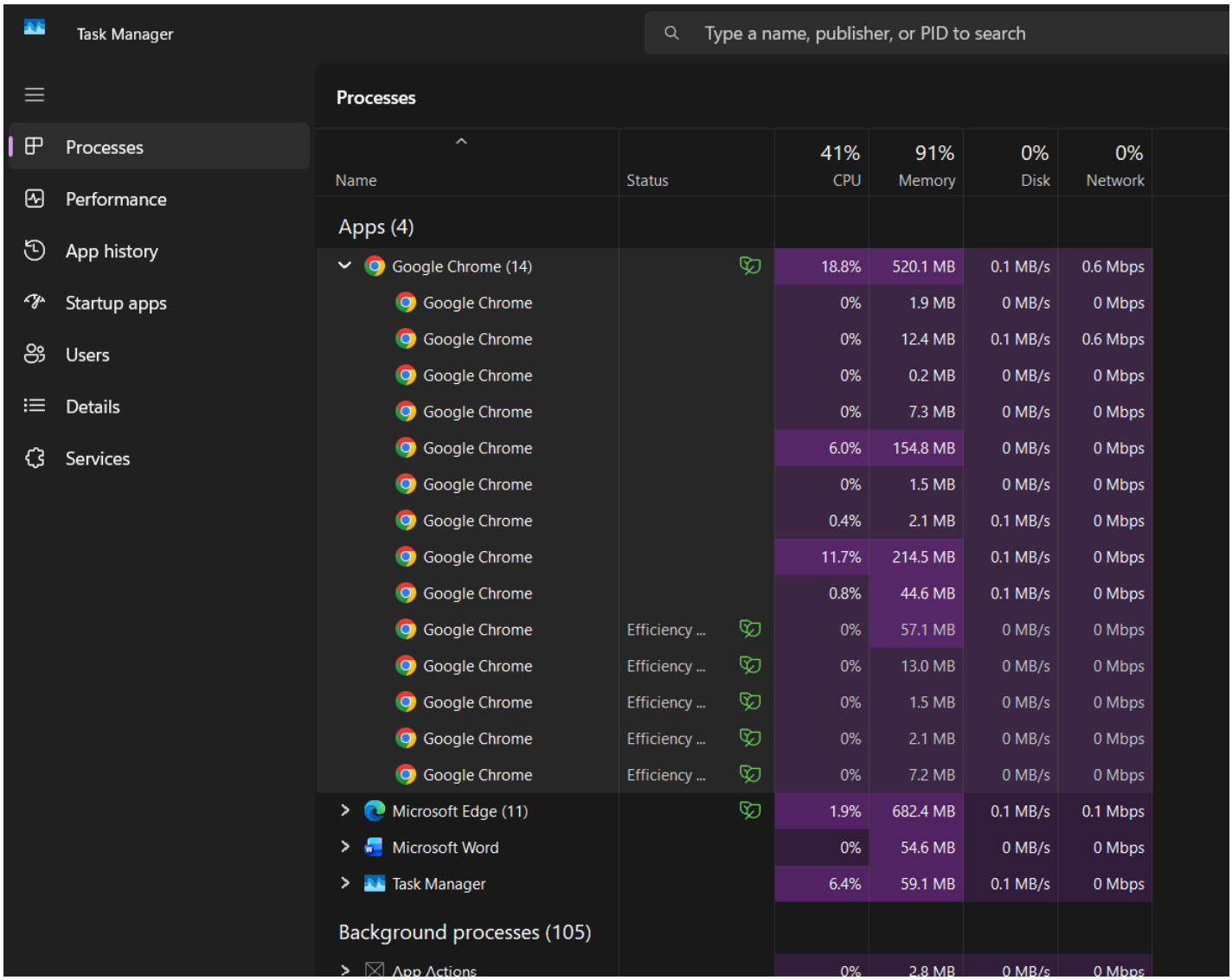
1. Press Ctrl + Shift + Esc to open Task Manager.
2. Select the **Processes** tab.
3. Observe the Network column to see the network activity of running applications.

➢ **Step 2: Open YouTube**

1. Open a web browser such as Google Chrome or Microsoft Edge.
2. Visit www.youtube.com.
3. Play any video and allow it to stream.

➢ **Step 3: Monitor Network Usage**

1. Keep the YouTube video playing for five minutes.
2. Open Task Manager and observe the Network column.
3. Check the network usage of the browser process.
4. Observe the network activity during the five-minute period.
5. Record the highest network usage value observed.



- **Step 4: Identify the Process**
  1. Observe the running processes in Task Manager.
  2. Identify which process shows the highest network activity during the monitoring period.
  3. Record its process name and network usage.

- **Observation**
- Record the actual values observed during the five-minute monitoring period.

Sr. No.	Observation	Recorded Value
1	Monitoring Duration	5 Minutes
2	Activity Performed	Streaming a YouTube Video
3	Peak Network Usage	0.6 Mbps
4	Process Using the Most Bandwidth	Google Chrome
5	Network Usage Unit	Mbps or Kbps

- **Explanation**
  1. **Network Usage:** Indicates the amount of data being transmitted or received by an application.
  2. **Peak Network Usage:** The highest network activity observed during the monitoring period.
  3. **Process:** A running application or background task that uses system resources.
  4. **Bandwidth:** The amount of data that can be transferred over a network within a given time.

➢ **Result**

➢ Network activity was monitored for five minutes while streaming a YouTube video using Windows Task Manager or Resource Monitor. The peak network usage and the process using the most bandwidth were identified and recorded.

❖ **4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets. Use the filter bar in Wireshark and try http or https as the filter keyword.**

➢ **Aim**

- To capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web using Wireshark, apply HTTP or HTTPS display filters, and identify three types of information visible in the captured packets.

➢ **Introduction**

- **Wireshark** is a network protocol analyzer used to capture and examine network packets travelling through a computer's network interface.
- When a user visits a website such as Flipkart or Instagram, the browser communicates with web servers using network protocols such as HTTP and HTTPS.

- **HTTP (Hypertext Transfer Protocol)** is used for transferring web resources, whereas **HTTPS (HTTP Secure)** provides encrypted communication using TLS.
- Wireshark allows users to capture network traffic and apply display filters to examine specific protocols. This practical helps us understand how web communication takes place.

➢ **Requirements**

- A Windows computer or laptop
- Internet connection
- Wireshark software
- Google Chrome or Microsoft Edge browser
- Flipkart or Instagram Web

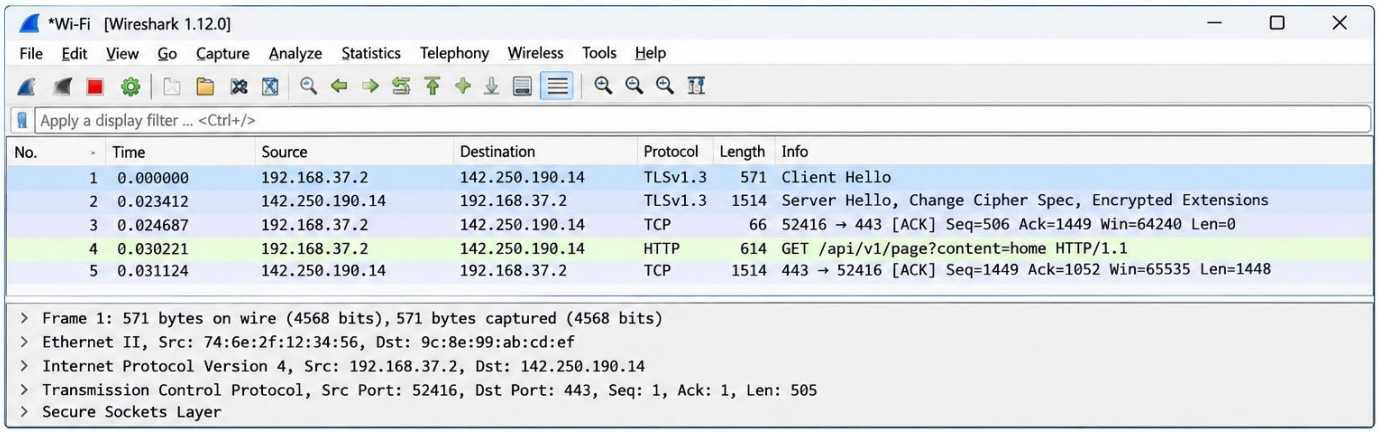
➢ **Procedure**

➢ **Step 1: Open Wireshark**

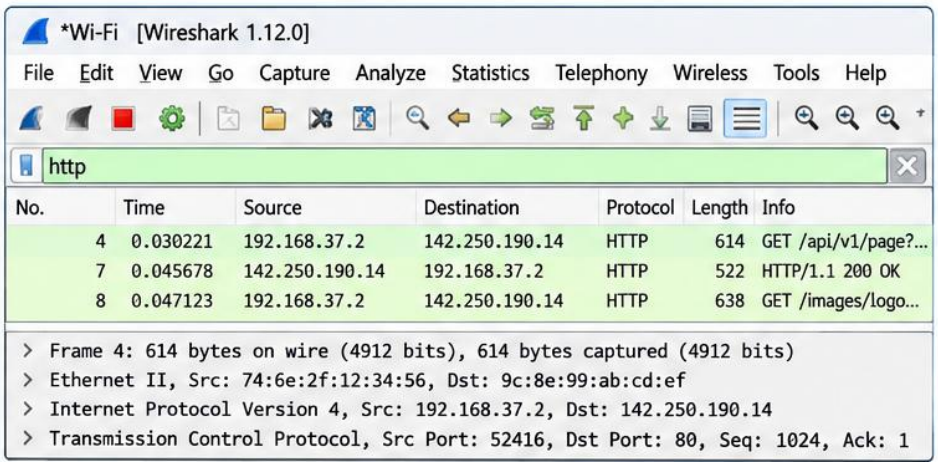
1. Open Wireshark on the laptop.
2. Identify the active network interface, such as Wi-Fi or Ethernet.
3. Double-click the active interface to start capturing packets.

➢ **Step 2: Browse a Website**

1. Open a web browser.
2. Visit https://www.flipkart.com or https://www.instagram.com.
3. Browse the website for at least 30 seconds.
4. Wireshark will capture the network packets generated during browsing.

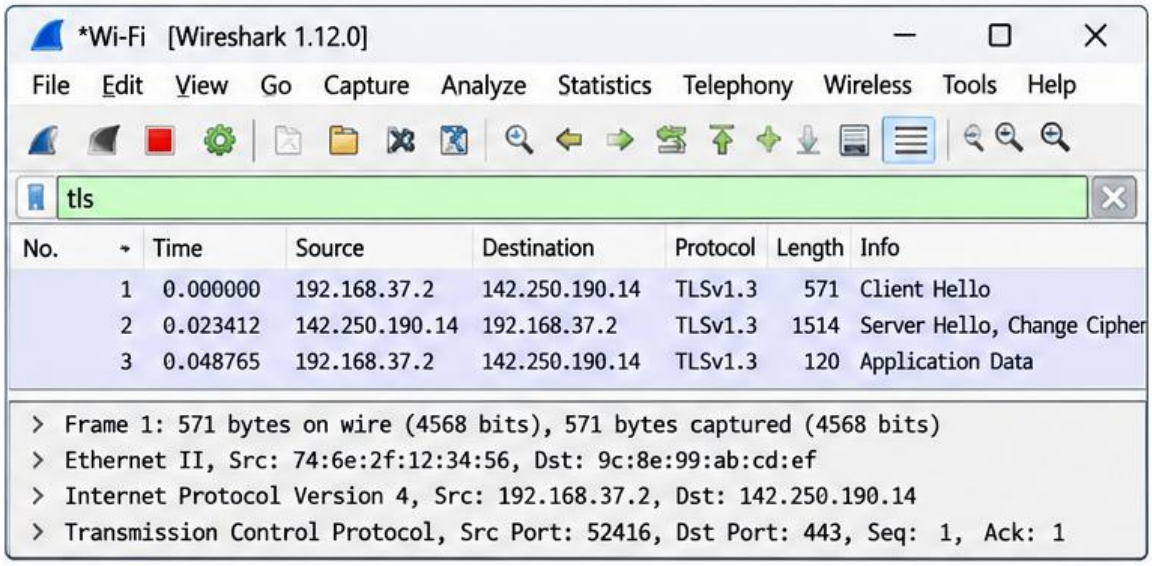


- **Step 3: Stop Packet Capture**
  1. Return to Wireshark.
  2. Click the red Stop button.
  3. The captured packets will be available for examination.
- **Step 4: Apply HTTP Display Filter**
  1. Locate the display filter bar at the top of Wireshark.
  2. Enter the following filter:
     http
  3. Press Enter to display packets identified as HTTP.



➢ **Step 5: Apply HTTPS-Related Filter**

1. Clear the previous filter.
2. Enter:
   tls
3. Press Enter to display TLS packets associated with HTTPS communication, if present.



➢ **Observation: Three Types of Information**

➢ The following three types of information can be observed in HTTP or HTTPS-related packets:

Information	Description
Source and Destination IP Addresses	Identify the source and destination devices or servers involved in network communication.
Port Numbers	Show the communication ports used by applications. HTTP commonly uses TCP port 80, while HTTPS commonly uses TCP port 443. HTTP/3 uses QUIC over UDP port 443.
Protocol and Packet Information	Show protocol details such as TCP, TLS, packet length, handshake information, and other available metadata.

- **Result**
- Network traffic was captured using Wireshark while browsing Flipkart or Instagram Web for at least 30 seconds. Display filters were applied to examine HTTP and HTTPS-related traffic, and three types of information were identified from the captured packets.

❖ **5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.**

➢ **Aim**

- To generate an AI-based troubleshooting checklist for improving home network speed, reducing lag, and improving reliability while playing online games.

➢ **Introduction**

- Slow internet, high ping, and packet loss can affect online gaming performance. These problems may cause lag, delays, and sudden disconnections.
- An AI tool such as ChatGPT can help identify common network problems and suggest practical ways to improve home network performance.

➢ **AI-Generated Troubleshooting Checklist**

➢ The following checklist was generated using ChatGPT to help troubleshoot slow internet while gaming online.

1) **Use a Wired Ethernet Connection:** Connect the gaming device directly to the router using an Ethernet cable. A wired connection is generally more stable and can reduce wireless interference and latency.

2) **Improve Router Placement:** Keep the Wi-Fi router in a central and open location, away from thick walls and electronic devices that may cause interference.
3) **Use the 5 GHz Wi-Fi Band:** If the gaming device is close enough to the router, connect to the 5 GHz Wi-Fi band for potentially faster speeds and less interference than the crowded 2.4 GHz band.
4) **Close Background Downloads:** Pause unnecessary downloads, software updates, video streaming, and other activities that may consume internet bandwidth while gaming.
5) **Restart the Router:** Restart the router if the connection becomes unstable. This may help resolve temporary network problems.
6) **Update Router Firmware:** Check the router manufacturer's instructions for firmware updates. Updates can improve performance, stability, and security.
7) **Check Ping and Packet Loss:** Use a suitable network diagnostic tool to check ping and packet loss. High ping or packet loss can cause lag even when the internet speed appears sufficient.
8) **Limit Unnecessary Connected Devices:** Disconnect devices that are not being used if they are consuming significant bandwidth.

- **Suggestion I Would Try First**
- **My first choice would be to use a wired Ethernet connection instead of Wi-Fi.**
- A wired connection provides a more stable link between the gaming device and the router. It can reduce wireless interference and help provide more consistent latency during online gaming.

- **Result**
- An AI-generated troubleshooting checklist containing eight suggestions was prepared to improve home network speed and reliability for online gaming. The first recommended step is to use a wired Ethernet connection.